

5 (a) Eukaryotic cells and prokaryotic cells contain DNA.

Complete the passage about DNA in eukaryotic cells and prokaryotic cells, using the most appropriate terms.

In eukaryotic cells, the DNA is located mainly in the chromosomes of the nucleus.

Chromosomal DNA is associated with proteins called Two other eukaryotic cell structures that contain DNA are mitochondria and

In prokaryotic cells, for example, the DNA is found in the and is usually circular.

[4]

- (b) Fig. 5.1 shows transcription of the first six nucleotides of a gene by the enzyme RNA polymerase. The bases of the first six nucleotides on DNA strand X are shown, but the bases on the template DNA strand are **not** shown.

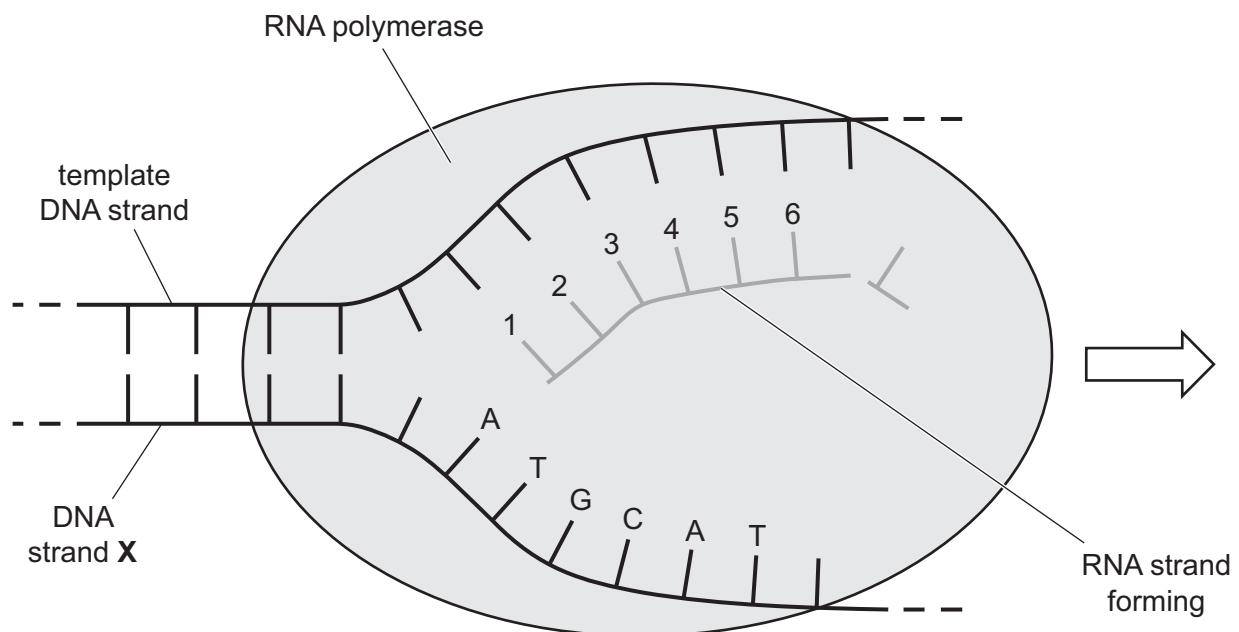


Fig. 5.1



(i) State the term used to describe the non-template DNA strand labelled **X** in Fig. 5.1.

..... [1]

(ii) Name the RNA strand formed during transcription of a eukaryotic gene.

..... [1]

(iii) Complete Table 5.1 to show the letters of the six bases indicated on Fig. 5.1 by the numbers 1 to 6.

Table 5.1

1	2	3	4	5	6

[1]

(c) RNA polymerase is composed of several polypeptides that move together and change shape as the enzyme performs its functions.

The death cap mushroom, *Amanita phalloides*, produces a toxin called alpha-amanitin, which binds to RNA polymerase at a site other than the active site. Alpha-amanitin reduces the activity of RNA polymerase.

Use the information provided to suggest how alpha-amanitin reduces the activity of RNA polymerase.

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..... [3]

[Total: 10]